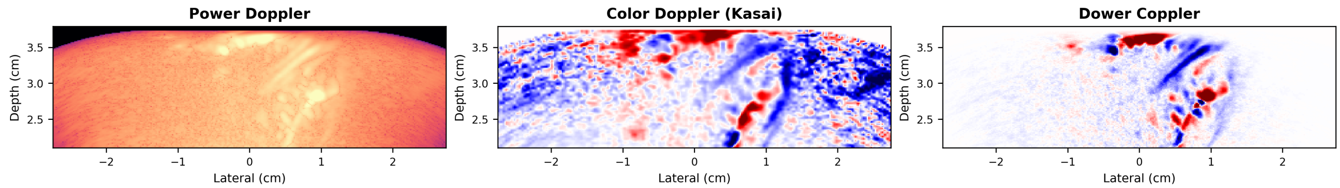
Dower Coppler: A Multi-Lag Coherence-Weighted Signed Doppler Index



Abstract

Ultrafast power Doppler imaging is sensitive to weak flow but discards flow direction, while classic autocorrelation color Doppler estimates signed velocity from phase and is sensitive to noise. We propose **Dower Coppler**, a composite signed Doppler index for SVD-filtered ultrafast compound data. The method does not introduce a new Doppler estimator from first principles; instead, it combines known building blocks—complex autocorrelation, multi-lag phase regression, power-like magnitude weighting, and fit-quality gating—into a single scalar image. For each pixel, we estimate the beam-projected phase velocity v_ϕ from a weighted multi-lag phase fit, then multiply it by the geometric mean of the lag-correlation magnitudes and by the weighted R^2 of the phase fit, yielding $v_\phi \, \mathrm{gmean}_k(|R_k|) \, R^2$. In Monte Carlo simulations across five noise and interference scenarios, the multi-lag phase estimator reduces median absolute frequency error by 2–3× relative to the Kasai estimator. We demonstrate the method *in vivo* on transcranial functional ultrasound data acquired from human cortex through the intact skull, without a cranial window, where the Dower Coppler index reveals opposing flow directions in adjacent vessels that are invisible in conventional power Doppler.

1 Introduction

Ultrafast ultrasound imaging, in which the entire field of view is insonified by unfocused or plane-wave transmissions at kilohertz frame rates, has enabled a new class of hemodynamic imaging modalities [1, 2]. Transcranial functional ultrasound (fUS) exploits this capability to image cerebral blood flow through the intact skull, but the transcranial setting imposes severe SNR constraints due to skull attenuation and aberration. Spatiotemporal SVD clutter filtering is now a standard preprocessing step in ultrafast Doppler and fUS because it separates tissue and blood components using their different spatiotemporal coherence [1]. After clutter filtering, the most common flow-strength image is power Doppler, which displays integrated Doppler power rather than mean Doppler frequency [3]. Power Doppler is robust and straightforward to compute, but it discards phase information: the resulting image is unsigned and carries no information about flow velocity or direction.

The Kasai autocorrelator [4] recovers a signed velocity estimate from the phase of the lag-1 autocorrelation, $\angle R_1$, and complex autocorrelation methods form the classical foundation of color flow imaging. Extensions using arbitrary spatial and temporal autocorrelation lags have also been studied in the color flow literature [5]. Thus, neither SVD clutter filtering, autocorrelation velocity estimation, nor the use of multiple lags is new by itself. The practical problem we address is how to combine signed phase information with a power-like confidence image for low-SNR transcranial ultrafast data.

Shen and Guo [6] introduced temporal multiply-and-sum (TMAS), the direct product-of-lag-magnitudes precursor to our coherence term, showing that multiplicative combinations of slow-time autocorrelations can improve power Doppler signal-to-noise ratio. Motivated by that idea, we use multi-lag autocorrelations to separate coherent flow from incoherent background. Rather than multiplying all lag magnitudes directly, we use their geometric mean as a bounded coherence term. This keeps the lag-product intuition—pixels must be temporally coherent across multiple lags—without the rapidly growing dynamic range of a raw product. On its own, however, any magnitude-only lag statistic inherits the fundamental limitation of power Doppler: it discards phase and produces an unsigned map.

A natural remedy is to use the autocorrelation phases, not just the magnitudes. For a coherent scatterer moving at angular frequency ω, the lag-k autocorrelation satisfies $\angle R_k \approx \omega k$: the phase is linear in lag. Fitting a line to $\{\angle R_k\}_{k=1}^{K}$ yields a multi-lag frequency estimate that uses all available lags rather than just lag 1. The idea of estimating frequency from the slope of autocorrelation phase is classical—Tretter [7] analyzed its statistical properties, and Fitz [8] developed fast approximations. Dower Coppler should therefore be viewed as a composite Doppler index, not a new physical Doppler estimator from first principles.

In this paper we propose **Dower Coppler**: a multi-lag, coherence-weighted signed Doppler index. It uses the signed axial phase velocity from a weighted multi-lag phase regression, multiplied by two confidence factors: the geometric mean of the lag-correlation magnitudes and the weighted R^2 of the phase-versus-lag fit. The regression weights are the autocorrelation magnitudes $|R_k|$, so that lags with stronger signal dominate the fitted phase slope, while pixels whose phase is not approximately linear in lag are attenuated by the fit-quality term. The distinctive contribution is this specific scalar product, $v_\phi \, G_R \, R^2$, for SVD-filtered ultrafast compound data.

The experimental setting is also relevant. Prior human fUS studies have primarily used intraoperative imaging after skull opening [9] or imaging through acoustically transparent cranial implants/windows [10, 11]. Here, the data are acquired transcranially through the intact adult skull. This extreme low-SNR operating point motivates a conservative composite index that uses multi-lag evidence and explicit phase-fit quality rather than a single-lag color Doppler estimate alone.

We evaluate the index and its phase-velocity component in two settings: (1) Monte Carlo simulations under additive white Gaussian noise, decorrelation, lag outliers, clutter leakage, and mixed-flow scenarios; and (2) *in vivo* transcranial functional ultrasound imaging of human cortex. Figure 1 previews the key result.

2 Methods

2.1 Signal Model

Let $s_t(\boldsymbol{x})$ denote the clutter-filtered complex analytic signal at pixel $\boldsymbol{x}$ and slow-time frame t, where $t = 0, 1, \dots, T-1$. For a coherent scatterer population moving at a constant Doppler angular frequency ω, the idealized signal is

$$s_t = A\, e^{j(\omega t + \phi_0)} + n_t, \tag{1}$$

where A is the (real, positive) amplitude, ϕ_0 is an arbitrary initial phase, and n_t is zero-mean additive noise.

The lag-k temporal autocorrelation is

$$R_k(\boldsymbol{x}) = \frac{1}{T-k} \sum_{t=0}^{T-k-1} s_{t+k}(\boldsymbol{x})\, s_t^*(\boldsymbol{x}). \tag{2}$$

In the noiseless, fully coherent case, $R_k \approx |A|^2 e^{j\omega k}$, so that

$$\angle R_k \approx \omega k. \tag{3}$$

2.2 Background: Multi-Lag Coherence

Standard power Doppler uses the zero-lag power $|R_0| = \frac{1}{T}\sum_t |s_t|^2$. A TMAS-style product of autocorrelation magnitudes across lags $k = 1, \dots, K$ is

$$M_{\text{TMAS}} = \prod_{k=1}^{K} |R_k|. \tag{4}$$

This construction is motivated by the TMAS product of autocorrelation magnitudes [6]. Because $|R_k|$ decays with lag for uncorrelated noise but remains roughly constant for coherent flow, the product suppresses background more strongly than $|R_0|$ alone. In the current Dower Coppler implementation, we use the geometric mean of the same lag magnitudes,

$$G_R = \exp\left(\frac{1}{K} \sum_{k=1}^{K} \log\left(\max(|R_k|, \epsilon)\right) \right), \tag{5}$$

with $\epsilon = 10^{-30}$ for numerical stability. This preserves the multi-lag coherence weighting while keeping the scale comparable to a single autocorrelation magnitude. Like the raw TMAS product, G_R is nonnegative and therefore carries no directional information.

2.3 Multi-Lag Phase Regression

We estimate the Doppler frequency from the slope of autocorrelation phase versus lag. Given the unwrapped phases

$$\phi_k = \mathrm{unwrap}\big(\angle R_k\big), \quad k = 1, \ldots, K, \tag{6}$$

we seek the angular frequency $\hat{\omega}$ that minimizes a weighted residual. The regression is constrained through the origin ($\phi_0 = 0$ at lag 0), since R_0 is real and positive after clutter filtering.

Plain weighted least squares (WLS). The initial estimate uses the autocorrelation magnitudes as weights:

$$\hat{\omega}_{\mathrm{WLS}} = \frac{\sum_{k=1}^{K} |R_k|\, k\, \phi_k}{\sum_{k=1}^{K} |R_k|\, k^2}. \tag{7}$$

This is the weighted regression of ϕ_k on k through the origin, where lags with stronger autocorrelation contribute more.

Fit quality. We quantify whether the phase sequence is well described by a single linear slope using a weighted coefficient of determination. With

$$\bar{\phi} = \frac{\sum_k |R_k|\, \phi_k}{\sum_k |R_k|}, \tag{8}$$

we compute

$$\mathrm{SSE} = \sum_k |R_k|\, (\phi_k - \hat{\omega}_{\mathrm{WLS}} k)^2, \tag{9}$$

$$\mathrm{SST} = \sum_k |R_k|\, (\phi_k - \bar{\phi})^2, \tag{10}$$

and

$$Q = R^2 = \mathrm{clamp}\left(1 - \frac{\mathrm{SSE}}{\max(\mathrm{SST}, 10^{-12})}, 0, 1\right). \tag{11}$$

Pixels with weak lag correlations or inconsistent phase evolution therefore receive low weight in the final image.

2.4 Dower Coppler

The Dower Coppler index combines the signed velocity estimated from the multi-lag phase slope with the multi-lag coherence and fit-quality terms. First, the phase slope in radians per slow-time frame is converted to a beam-projected axial phase velocity:

$$v_\phi = \hat{\omega}_{\mathrm{WLS}} \frac{f_{\mathrm{slow}}}{2\pi} \frac{c}{2 f_0}, \tag{12}$$

where f_{slow} is the cadence of the slow-time samples, c is the assumed speed of sound, and f_0 is the transmit center frequency. For compounded plane-wave data, f_{slow} is the transmit pulse repetition frequency divided by the number of transmissions per compounded frame. This is the standard axial Doppler conversion; no beam-flow angle correction is applied. The final image is

$$I_{\mathrm{Dower}} = v_\phi\, G_R\, Q. \tag{13}$$

The result is positive for one axial flow direction and negative for the opposite direction, depending on the system sign convention. The sign comes from v_ϕ, while G_R and Q attenuate pixels with weak temporal coherence or poor phase linearity. Because G_R is signal-scale dependent, I_{Dower} should be interpreted as a signed Doppler evidence index rather than a quantitative velocity map; v_ϕ is retained separately when velocity is the quantity of interest.

The per-pixel computational cost is small: the autocorrelations R_k are computed for K lags, followed by closed-form weighted least squares and a weighted R^2 calculation.

2.5 Phase Unwrapping

The autocorrelation phases $\angle R_k$ are individually wrapped to $(-\pi, \pi]$. Before regression, we apply a one-dimensional temporal unwrap along the lag axis: for each consecutive pair of lags, the phase difference is corrected to lie within $(-\pi, \pi]$ by adding integer multiples of 2π. This is a standard 1-D unwrap and does not require spatial information.

The final R^2 quality term attenuates isolated unwrapping failures: a miswrapped lag produces a large residual from the linear phase model, lowering the weighted fit quality for that pixel.

3 Simulation Study

3.1 Setup

We evaluate the phase slope estimators on synthetic slow-time ensembles of $T = 64$ frames. For each trial, a coherent signal is generated according to eq. (1) with unit amplitude and additive complex Gaussian noise at a specified SNR. The true angular frequency ω is drawn from the set $\{0.35, 0.8, 1.35\}$ rad/frame, and the SNR ranges from -12 to $+12$ dB. We compute lag autocorrelations up to $K = 5$ and run $N = 1024$ independent trials per condition.

Five scenarios are tested:

1. **AWGN**: coherent flow plus white Gaussian noise.
2. **Decorrelation**: flow with a slow phase random walk (models scatterer turnover).
3. **Lag outlier**: one high-lag autocorrelation is corrupted by a large additive phase error.
4. **Clutter leakage**: a residual slow-moving clutter component is added to the flow signal.
5. **Two-component flow**: a weaker flow component with the opposite sign is mixed with the primary flow.

3.2 Estimators Compared

- **Kasai**: $\hat{\omega} = \angle R_1$ (single-lag baseline).
- **WLS**: weighted least-squares phase slope (eq. (7)).
- **Huber-WLS**: robust comparison estimator using Huber residual weights ($\delta = 0.7$, 5 iterations).
- **Circular-sign-WLS**: direction from a circular objective $\arg\max_\omega \sum_k |R_k| \cos(\phi_k - \omega k)$, magnitude from WLS.

3.3 Metrics

For each estimator, we report:

- **Median absolute error** (rad): median of $|\hat{\omega} - \omega|$.
- **RMSE** (rad): root mean squared error of the frequency estimate.
- **Sign error rate**: fraction of trials where $\mathrm{sgn}(\hat{\omega}) \neq \mathrm{sgn}(\omega)$.

4 In Vivo Experiment

4.1 Ethics Approval

Human imaging was performed under WCG IRB Protocol #20254791. The volunteer provided informed consent prior to data acquisition. Ultrasound acquisition was performed within the approved IRB safety limits.

4.2 Data Acquisition

Transcranial functional ultrasound data were acquired from the temporal lobe of one healthy volunteer using a Butterfly iQ+ point-of-care ultrasound probe operated with custom firmware. The probe was placed over the temporal acoustic window and the acquisition was performed through the intact skull, without craniotomy, cranioplasty, or another acoustic cranial window. The source ultratrace contained 480 acquisitions. Each acquisition contained 700 compounded slow-time frames formed from five azimuthal plane-wave transmissions spanning $\pm 6°$; two additional noise loops were stored in the raw IQ data but were not used for Doppler processing. The transmit center frequency was 2.75 MHz with three transmit cycles, and the receive data were sampled at 4.17 MHz after decimation. The HDF5 metadata report an empirical pulse-repetition rate of 1245.9 Hz over the acquisition subset used for the main figures; because each compounded frame used five transmit angles, the corresponding compounded-frame cadence was approximately 249 Hz. The assumed sound speed was $1600\,\mathrm{m\,s^{-1}}$.

The original beamformed volume covered -27.456 mm to 27.456 mm laterally, -20 mm to 20 mm elevationally, and 21.0 mm to 37.896 mm in depth on a $147 \times 30 \times 58$ lateral-elevation-depth grid. The main paper figures were generated from a fine lateral/depth rebeamform of the two central elevation planes (original elevation indices 14 and 15), using acquisitions 200–399. This figure dataset used a $294 \times 2 \times 116$ lateral-elevation-depth grid over the same lateral and depth extent. The temporal-stability analysis used precomputed per-buffer Doppler arrays and explicitly varied the number of buffers included in the median.

Table 1: Key acquisition and reconstruction parameters for the transcranial human fUS dataset.

Parameter	Value
Target / window	Temporal lobe / temporal acoustic window
Probe	Butterfly iQ+ with custom firmware
Acquisitions in ultratrace	480
Frames per acquisition	700 compounded slow-time frames
Transmit angles	5 azimuthal angles over $\pm 6°$
Transmit frequency	2.75 MHz, 3 cycles
Empirical pulse PRF	1245.9 Hz (acq. 200–399 mean)
Compounded-frame cadence	249 Hz
Sampling rate	4.17 MHz after decimation
Depth range	21.0 mm to 37.896 mm
Sound speed	$1600\,\mathrm{m\,s^{-1}}$
Original grid	$147 \times 30 \times 58$ (lat. $\times$ elev. $\times$ depth)
Figure grid	$294 \times 2 \times 116$ (lat. $\times$ elev. $\times$ depth)

4.3 Processing Pipeline

Clutter filtering was performed by singular value decomposition (SVD) of the slow-time ensemble at each pixel, with low-rank tissue clutter components removed. In the current processing, the compound data are mean-subtracted before SVD filtering; the lowest 8% of singular modes are removed, the high cutoff is 1.0, the fast SVD implementation is used, and the first five filtered frames are omitted from Doppler summaries and phase fitting. Beamforming used the MACH GPU delay-and-sum backend, channel mean subtraction, an f-number of 0.5, all enabled receive rows, and one-row compounding. The source ultratrace was beamformed into 30 elevation planes with 20 processing chunks; the fine-grid figure dataset used the two central elevation planes at doubled lateral/depth sampling. All Doppler estimators share the same filtered ensemble; they differ only in how the filtered slow-time data are reduced to an image. We computed:

- Standard power Doppler ($|R_0|$).
- Kasai color Doppler ($\angle R_1$).
- Multi-lag phase velocity from weighted phase regression.
- Dower Coppler, $v_\phi\, G_R\, R^2$, at $K = 5$.

5 Results

5.1 Simulation Results

Table 2 summarizes the estimator performance across all five scenarios.

Across all scenarios, WLS and Huber-WLS reduce the median absolute frequency error by approximately 2–3$\times$ relative to the Kasai estimator. In the clean AWGN case, WLS and Huber-WLS are nearly identical, confirming that the robust reweighting does not degrade performance when no outliers are present. Under the lag-outlier scenario, Huber-WLS achieves the lowest median error and RMSE, as the Huber weights suppress the corrupted lag. The circular-sign-WLS hybrid consistently achieves the lowest sign error rate, particularly under clutter leakage and two-component flow.

5.2 In Vivo Results

As shown in fig. 1, power Doppler delineates the cortical vasculature with high sensitivity but provides no directional information. Kasai color Doppler (center) reveals flow direction but is dominated by noise, with vessel structure largely obscured. The proposed Dower Coppler (right) attenuates incoherent pixels using the multi-lag magnitude and fit-quality terms while recovering flow direction from the multi-lag phase slope. Adjacent cortical vessels with opposing flow directions—invisible in power Doppler—are clearly separated in the Dower Coppler image.

Figure 2 isolates the contribution of the main terms in the composite index. The phase velocity v_ϕ alone contains the directional pattern but also substantial background phase noise. Multiplication by the multi-lag coherence strength G_R suppresses decorrelated background and emphasizes vessel-like

Table 2: Estimator performance averaged over SNR and velocity conditions. Best values in each scenario are shown in **bold**.

Scenario	Estimator	Med. abs. err. (rad)	RMSE (rad)	Sign err. (%)
AWGN	Kasai	0.235	0.401	8.2
	WLS	**0.095**	0.257	6.8
	Huber-WLS	0.096	0.259	6.8
	Circ-WLS	0.101	**0.253**	**6.5**
Decorrelation	Kasai	0.243	0.415	8.4
	WLS	**0.112**	0.277	7.1
	Huber-WLS	**0.112**	0.278	7.0
	Circ-WLS	0.125	**0.274**	**6.8**
Lag outlier	Kasai	0.241	0.401	8.6
	WLS	0.094	0.276	6.5
	Huber-WLS	**0.092**	**0.267**	6.6
	Circ-WLS	0.105	0.271	**6.4**
Clutter leak	Kasai	0.380	0.546	10.0
	WLS	0.182	0.373	9.8
	Huber-WLS	0.180	0.374	9.6
	Circ-WLS	**0.178**	**0.368**	**8.5**
Two-component	Kasai	0.338	0.518	10.7
	WLS	**0.128**	0.315	9.0
	Huber-WLS	**0.128**	0.316	8.9
	Circ-WLS	0.133	**0.308**	**8.3**

structures. Multiplication by R^2 preserves the signed velocity interpretation while down-weighting pixels whose lag phases are poorly explained by a single slope. The G_R panel shows that the magnitude term is power-Doppler-like and unsigned; direction enters through v_ϕ.

Figure 3 separates the signed velocity estimate from the Dower Coppler amplitude weighting. The multi-lag phase velocity preserves the dominant flow-direction pattern seen by the lag-1 Kasai estimator, but the Bland–Altman comparison shows that the two velocity estimates are not interchangeable at high-magnitude pixels. In this dataset the mean bias is near zero, while the limits of agreement are set by pixels where the lag-1 phase and the multi-lag phase-slope fit disagree.

Figure 5 quantifies this comparison using contrast-to-noise ratio (CNR) and generalized CNR (gCNR) across six vessel ROIs exported from the Doppler CNR viewer. For each exported signal mask, we use the largest circular ROI fully contained inside that mask. The same vessel-free background rectangle is used for every signal ROI so that the CNR values differ only because of the selected vessel pixels and Doppler image.

Dower Coppler achieves median CNR 21.5 dB across the six largest inscribed circular vessel selections, with values ranging from 11.2–40.1 dB. Power Doppler has median CNR 14.7 dB (range 9.9–19.3 dB), and Kasai color Doppler has median CNR -7.1 dB (range -32.7–3.9 dB) when the signed Doppler maps are measured by magnitude, as in the viewer CNR calculation. The upper end of the Dower Coppler range is driven by V6, a one-pixel inscribed circle on a compact high-contrast vessel; we therefore emphasize the median as the more representative summary statistic. The gCNR metric confirms that the selected vessels are well separated from the shared background region for all three displays, with Dower Coppler and power Doppler at approximately 1.0 across all selections.

As a repeatability check, we split the September 21 acquisitions into two non-overlapping halves (200–299 and 300–399), recomputed median Dower Coppler maps from the per-acquisition fine-elevation sidecar, and compared signs in the same circular vessel ROIs on the closest available elevation plane ($y = -0.44$ mm). The Dower Coppler sign was iden-

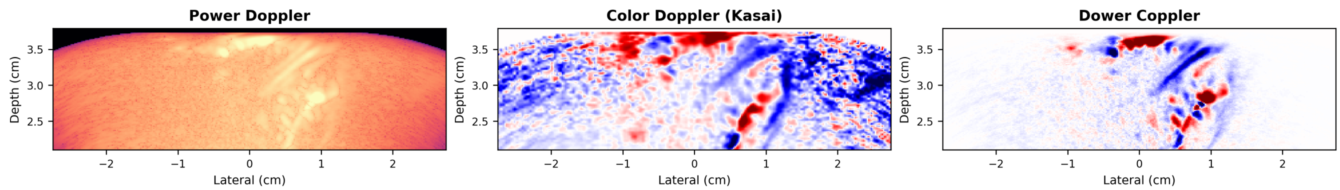


Figure 1: Transcranial cortex imaging (median of 480 Doppler buffers, plane-wave compounding). **Left**: power Doppler, displayed from −100 to 140 dB, shows vessel structure but no directional information. **Center**: Kasai color Doppler ($\angle R_1$) reveals direction but is dominated by noise. **Right**: Dower Coppler ($v_\phi\, G_R\, R^2$, $K = 5$) combines multi-lag coherence weighting with directional information from the phase slope and fit quality. Opposing flow directions in adjacent cortical vessels are clearly resolved.

tical between halves for all 130 vessel-ROI pixels.

Figure 6 demonstrates the temporal stability of the Dower Coppler map by varying the number of Doppler buffers averaged.

The vessel pattern and flow direction assignments are stable from 250 buffers down to as few as 5 buffers. At a single buffer, the major vessels remain identifiable but background noise increases substantially.

Figure 7 shows the Dower Coppler map across elevation planes 2–7 from a finer elevational rebeamforming run spanning $y = -4$ to 4 mm, providing a qualitative check that the directional vascular pattern is not confined to a single selected plane.

Figure 8 shows the same Dower Coppler processing on a separate recording, included as a qualitative check that the displayed vascular contrast is not unique to the September 21 dataset. The figure-generation script exposes the source NPZ path and elevation plane as command-line parameters so this example can be replaced without editing the paper.

6 Discussion

Dower Coppler is best viewed as a composite Doppler index rather than a new Doppler estimator from first principles. Its ingredients—SVD clutter filtering, complex autocorrelation Doppler, multi-lag phase estimation, and power-like magnitude weighting—are well established. The distinct design choice is the scalar product $v_\phi\, G_R\, R^2$: signed phase velocity multiplied by multi-lag geometric coherence strength and by phase-linearity quality. This hybrid output is signed like color Doppler, amplitude-weighted like power Doppler, and confidence-gated by the quality of a single multi-lag phase-slope model.

Relationship to prior work. SVD filtering before ultrafast Doppler is standard in fUS and related microvascular imaging [1]. Autocorrelation phase velocity estimation traces directly to Kasai color Doppler [4], and arbitrary temporal/spatial lags were analyzed in classic color-flow autocorrelation work [5]. The Loupas two-dimensional autocorrelator also uses temporal and depth lags jointly, but for axial frequency correction rather than multi-lag phase-slope fitting [12]. Multi-lag autocorrelation has deep roots in weather radar, where related estimators fit correlation models across multiple lags rather than fitting a Doppler phase line [13, 14]. Frequency estimation from phase regression is also classical [7, 8]. Power Doppler is a close conceptual neighbor because it encodes flow strength rather than pure velocity [3]. The TMAS work of Shen and Guo is especially relevant because it directly uses products of temporal autocorrelation magnitudes to improve ultrafast power Doppler SNR [6]; G_R can be interpreted as a bounded geometric-mean version of this product-of-lag-magnitudes idea. Relative to these methods, the contribution here is the specific coherence-weighted signed index, and in particular the use of a per-pixel phase-linearity R^2 as a confidence gate. This R^2 term differs from standard spectral-width or variance estimates: it measures how well the lag phases are explained by one linear slope, penalizing multi-component flow, turbulence, wrapping errors, and residual clutter. The claim is therefore not novelty for autocorrelation Doppler, SVD filtering, or multi-lag correlation processing broadly.

Choice of parameters. The maximum lag $K = 5$ was chosen empirically: higher lags provide more fitting leverage but also more decorrelation, particularly through the skull. We used an SVD low cut-

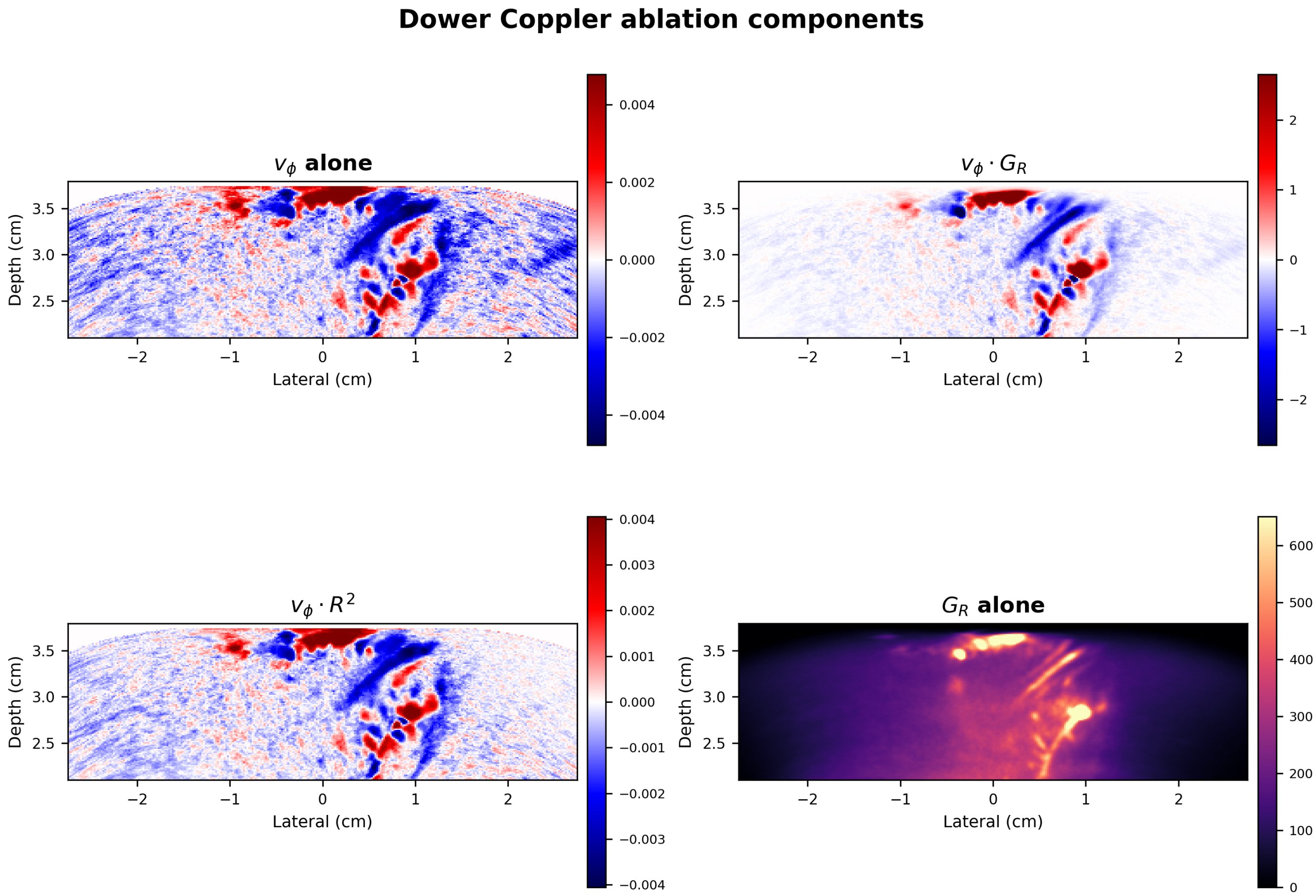


Figure 2: Ablation of the Dower Coppler components on the same transcranial image plane. v_ϕ alone is the signed multi-lag phase-velocity estimate. $v_\phi G_R$ adds the geometric mean of multi-lag autocorrelation magnitudes, suppressing incoherent background. $v_\phi R^2$ gates the signed velocity by the phase-linearity fit quality. G_R alone is the unsigned coherence-strength image. The final Dower Coppler image uses $v_\phi G_R R^2$.

off of 0.08, high cutoff of 1.0, mean subtraction before SVD, and omitted the first five filtered frames from Doppler estimation. The current implementation uses weighted least squares rather than Huber reweighting for the final Dower Coppler image; Huber-derived quality maps remain useful diagnostics but are not part of the default formula.

Limitations. This study uses data from a single volunteer with a single probe. The transcranial imaging scenario—with low SNR, strong clutter, and skull-induced aberrations—is a challenging test case, but validation on a larger cohort and with flow phantoms providing ground-truth velocity fields is needed. Dower Coppler is not itself a quantitative velocity image: G_R depends on gain, depth attenuation, beamforming normalization, SVD scaling, and backscatter strength. Quantitative velocity analysis should use v_ϕ directly, with appropriate attention to angle dependence and aliasing. The conversion in eq. (12) estimates the beam-projected axial component; it does not correct for beam-flow angle. Multi-lag fitting also does not remove the fundamental pulse-Doppler phase-wrap ambiguity: the lag phases still have to be unwrap-able. Finally, the method assumes a single dominant flow direction per pixel; bidirectional or turbulent flow can produce a compromised single-slope estimate, though the R^2 term attenuates pixels whose phase sequence is poorly described by a single velocity.

Extensions. The fit quality metric Q is already used as a multiplicative confidence term, but it can also be displayed as a standalone quality map. Robust alternatives, such as Huber-weighted slopes or circular-sign hybrids, remain useful variants for ap-

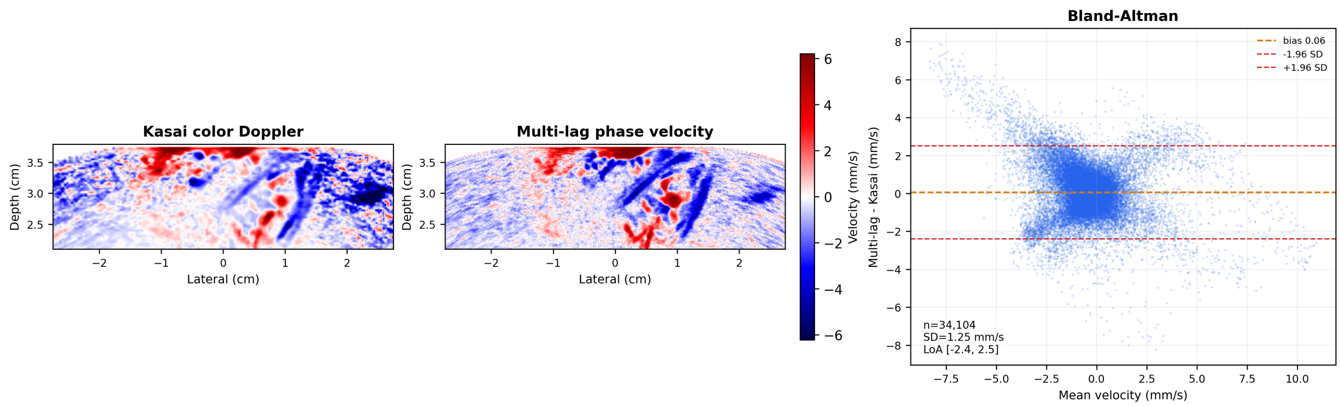


Figure 3: Comparison of conventional lag-1 Kasai color Doppler and the multi-lag phase-velocity estimate used by Dower Coppler. **Left**: Kasai velocity, computed from $\angle R_1$. **Center**: multi-lag phase velocity v_ϕ from the weighted phase-slope fit. Both velocity maps are shown in mm/s with the same symmetric color scale. **Right**: Bland–Altman plot comparing the two velocity estimates over all finite pixels in the displayed plane.

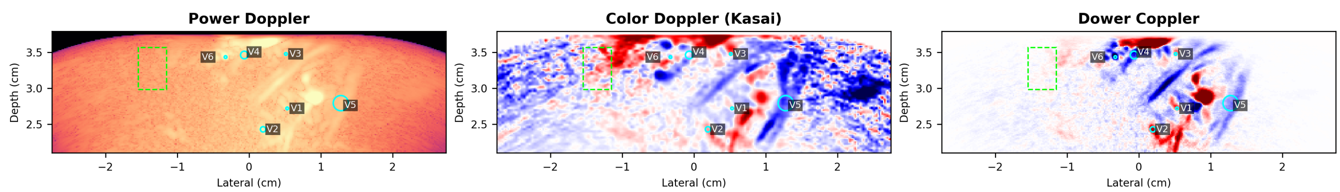


Figure 4: Regions used for the CNR and gCNR measurements in fig. 5. Cyan circles indicate the largest circular signal ROIs fully contained inside the six exported vessel masks. The dashed green rectangle is the shared background ROI used for every vessel and every Doppler display.

plications where isolated lag outliers or sign accuracy dominate the design objective. Future ablations should explicitly compare power Doppler, Kasai color Doppler, directional power variants, first-lag magnitude, multi-lag phase velocity alone, velocity times G_R, velocity times R^2, and the full $v_\phi G_R R^2$ index.

7 Conclusion

We have proposed Dower Coppler, a multi-lag coherence-weighted signed Doppler index for SVD-filtered ultrafast compound ultrasound. The method estimates signed beam-projected phase velocity from a multi-lag autocorrelation phase fit and weights it by lag-correlation strength and phase-fit quality. It adds signed directional contrast at minimal computational cost while attenuating weak or inconsistent pixels. Simulations show that multi-lag phase regression reduces frequency estimation error by 2–3× relative to the single-lag Kasai estimator. Transcranial *in vivo* imaging demonstrates that the method resolves opposing flow directions in adjacent cortical vessels—information that is lost in conventional power Doppler.

Code, reproducible analysis scripts, and the derived NPZ inputs used for the manuscript figures are available in the standalone paper repository: github.com/ennucore/dower-coppler.

References

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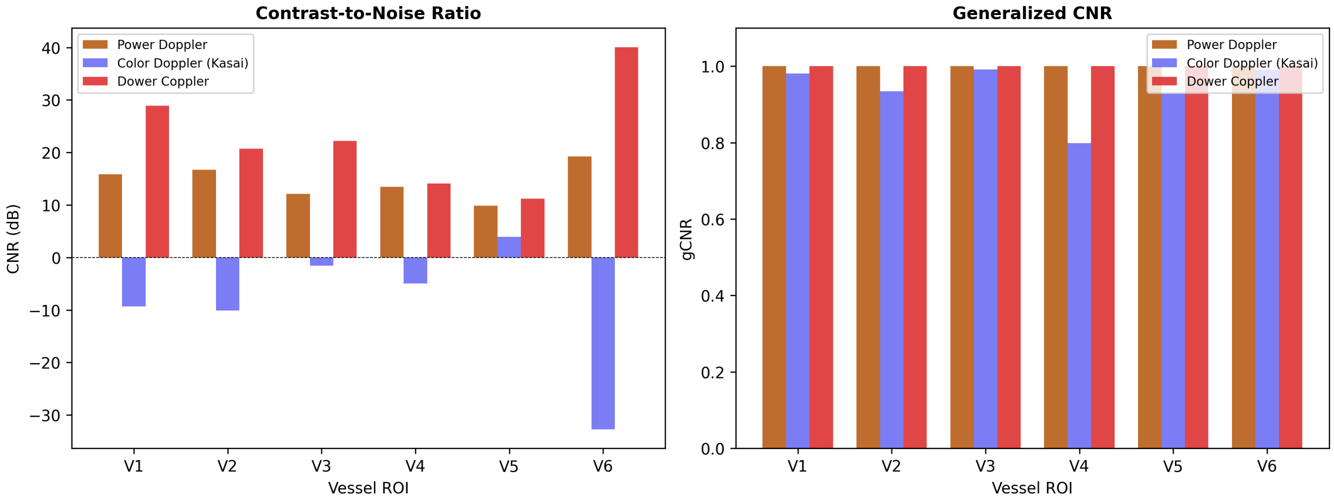

Figure 5: **Left**: CNR in dB across six vessel ROIs. Dower Coppler has median CNR 21.5 dB across the six selections (range 11.2–40.1 dB), exceeding both power Doppler and Kasai color Doppler in all ROIs. **Right**: generalized CNR (1 − histogram overlap). All methods show high gCNR for the selected vessel/background separation, with Dower Coppler and power Doppler close to 1.0 across all selections.

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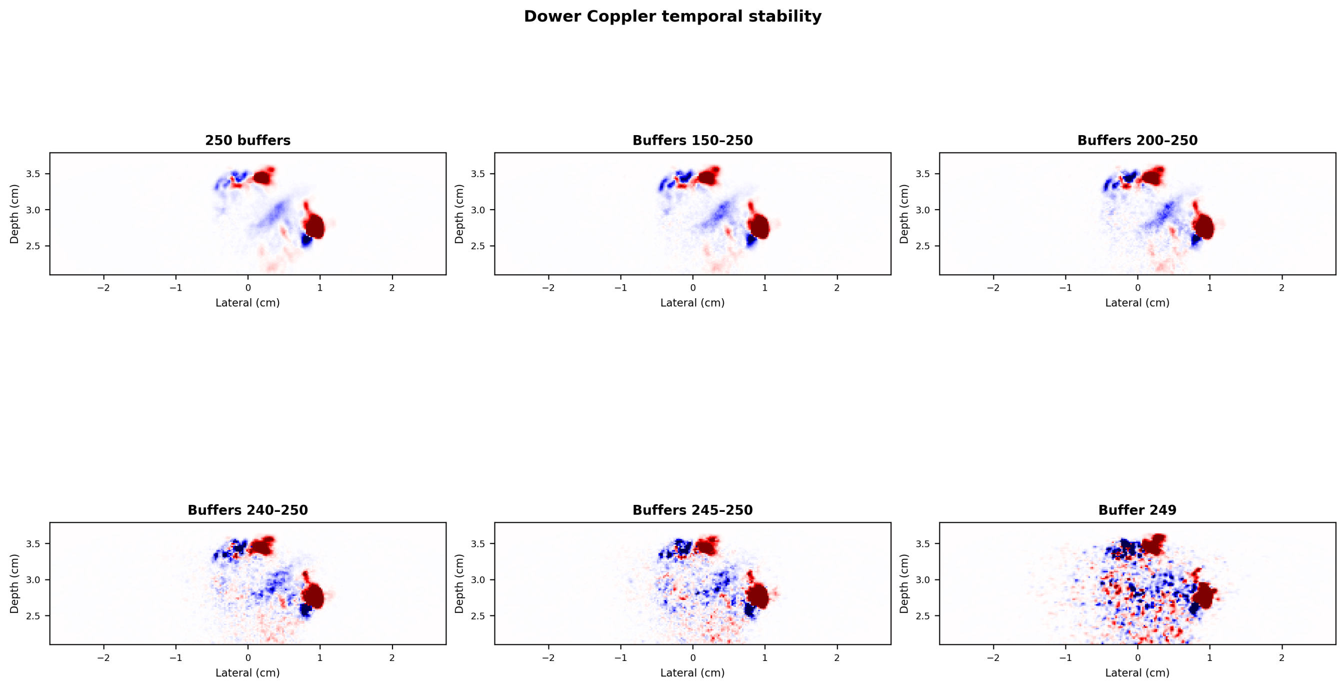

Figure 6: Dower Coppler maps computed from different numbers of Doppler buffers (same color scale throughout). The vessel pattern and flow direction assignment are stable from 250 buffers down to 5 buffers. At a single buffer, the major vessels remain identifiable but background noise increases.

Generowicz, Djaina D. Satoer, Clemens M. F. Dirven, Marion Smits, Borbála Hunyadi, Sebastiaan K. E. Koekkoek, Arnaud J. P. Vincent, Chris I. De Zeeuw, and Pieter Kruizinga. Mobile human brain imaging using functional ultrasound. *Science Advances*, 11(25):eadu9133, 2025. doi: 10.1126/sciadv.adu9133.

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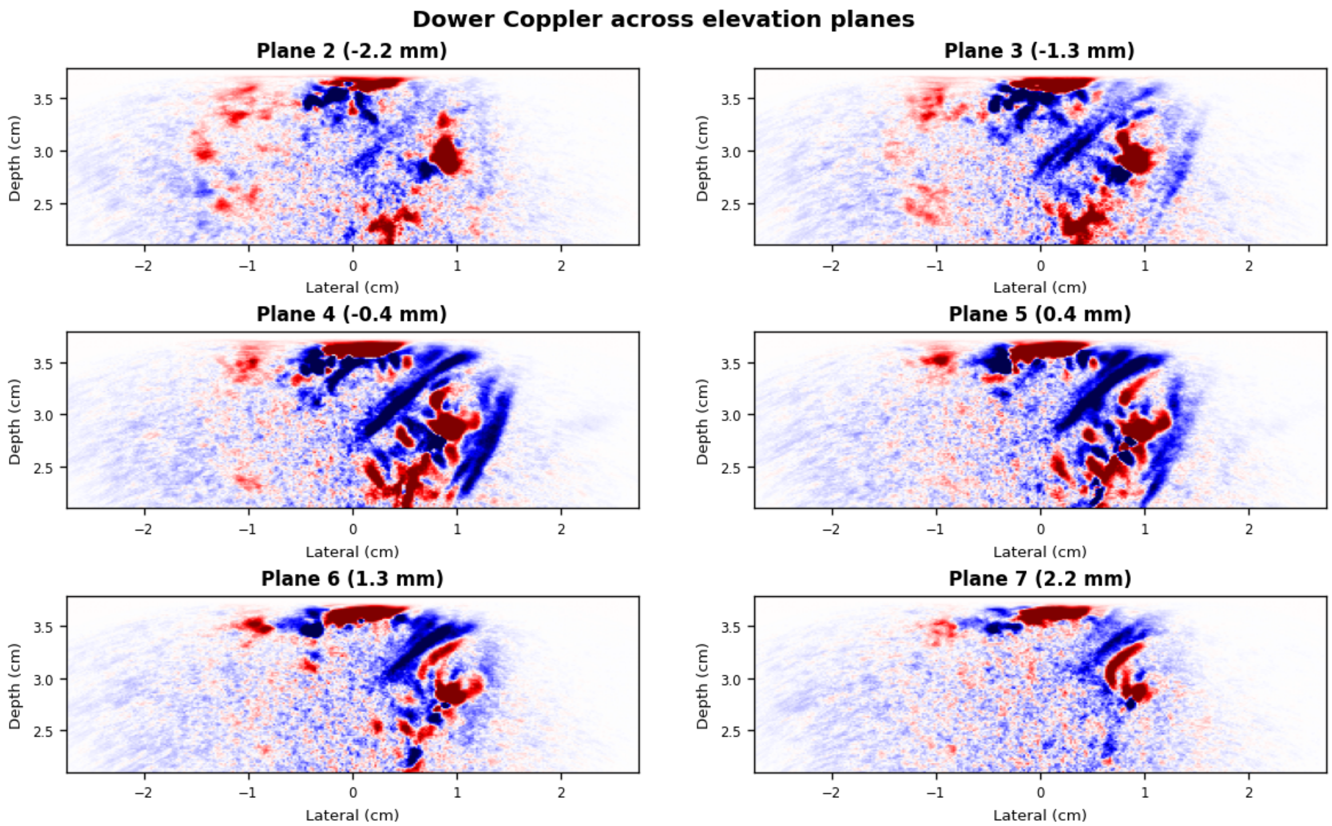

Figure 7: Dower Coppler maps across elevation planes 2–7 from the finer elevational rebeamforming run spanning $y = -4$ to 4 mm with 10 elevation planes. The panels use acquisitions 200–400 and one shared symmetric color scale computed from the 97th percentile of absolute Dower Coppler values across the displayed planes. The directional vascular pattern appears across the imaging volume, with similar vessel segments visible in adjacent planes.

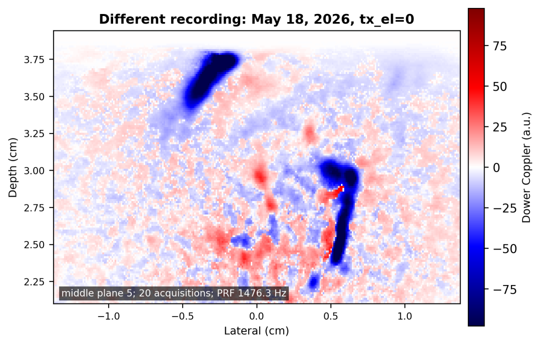

Figure 8: Dower Coppler image from a different recording: May 18, 2026, BT24480388, transmit element 0. The displayed image is the middle plane from a 10-plane GPU/MACH beamformed H5 ultratrace using row_index=-1, $y = -3.5$ to 3.5 mm, fine lateral/axial coarseness 0.5/0.5, 20 acquisitions (0–19), median aggregation, PRF 1476.3 Hz, $f_0 = 2.75$ MHz, $c = 1600$ m/s, SVD cutoff 0.08, and $K = 5$. The main September 21 dataset uses a separate Head ultratrace, acquisitions 200–399 for the CNR/fine-grid analysis, elevation planes 2–7 from the $y = -4$ to 4 mm fine-elevation run for the volume montage, pulse PRF 1244.2 Hz (compounded-frame cadence 248.8 Hz), and the same f_0, c, SVD cutoff, and K.